



September 2019 Examination

EC400

Probability and Statistical Inference

Instructions to candidates

Time allowed: 2 hours

This paper contains **SEVEN** questions. Answer **ALL** questions.

On your script, write

- Your name
- Your class group number
- The name of your class teacher.

You are not allowed to use an electronic calculator

1. **(8 points - 10 minutes)** I brought various bags with tulip bulbs from the Netherlands for my friends. Each bag contains 25 tulip bulbs. Three-fourths of the bags contain bulbs for 5 red and 20 yellow tulips; one fourth of the bags contain bulbs for 15 red and 10 yellow tulips. A bag is selected at random and one bulb is planted. Give the probability that it will produce a

- (a) Red tulip, say $P(R)$.
- (b) If the tulip is red, find the conditional probability that a bag having 15 red and 10 yellow tulips was selected.

2. **(8 points - 10 minutes)** Let X have the p.d.f. $f(x) = x^2/9$, $0 < x < 3$, zero elsewhere. Find the p.d.f. of $Y = X^3$. What is the median of the distribution of Y ?

3. **(13 points - 15 minutes)** The joint probability density function of X and Y is given by

$$f(x, y) = 2e^{-x-y}, 0 \leq y \leq x < \infty$$

- (a) Find the marginal p.d.f. of X .
- (b) Find the conditional p.d.f. of Y given $X = x$.
- (c) Are X and Y independent?

4. **(12 points - 14 minutes)** Let X_1 and X_2 be continuous random variables. If X_1 and X_2 are independent it can be shown that $E(g(X_1)h(X_2)) = E(g(X_1))E(h(X_2))$ where g and h are continuous functions.

- (a) Discuss how we can obtain $E(g(X_1)h(X_2))$ in the setting where X_1 and X_2 are not independent using the joint density of X_1 and X_2 .
- (b) An alternative method to obtain this expectation uses the law of iterated expectations. In view of this, show how you can obtain this expectation using the conditional distribution $f_{X_2|X_1=x_1}$ and the marginal distribution f_{X_1} instead. Clearly explain the intuition behind the law of iterated expectations and prove its validity in this setting.

5. **(25 points - 30 minutes)** The random variable X is normally distributed with mean μ and variance $\sigma^2 = 100$. Using the sample average, $\bar{X}$, we want to test the null hypothesis $H_0 : \mu = 10$ against the alternative $H_1 : \mu \neq 10$ at the 5% level of significance. A sample of $n = 100$ independent realizations of X was collected.

(a) Discuss how you can obtain the probability of committing a type-II error if the true value of μ is 10.5.

i. What happens to this probability if the variance σ^2 becomes smaller and/or the sample n becomes bigger? Discuss.

ii. What happens to this probability if you had considered the one-sided alternative $H'_1 : \mu > 10$ at the 5% level of significance instead? Discuss.

(b) What is the p-value of the test if $\bar{x} = 10.22$?

i. What happens to this p-value if the variance σ^2 becomes smaller and/or the sample n becomes bigger? Discuss.

ii. What happens to this p-value if you had considered the one-sided alternative $H'_1 : \mu > 10$, instead? Discuss.

6. **(12 points - 15 minutes)** The random variable X is said to have a $\chi^2(r)$ distribution. Its moment generating function is given by

$$M(t) = (1 - 2t)^{-r/2} \text{ with } t < 1/2.$$

(a) Use the moment generating function to show that the mean and variance of the random variable X equal to r and $2r$ respectively.

(b) Based on a random sample $X_1, \dots, X_n$, provide two method of moment estimators for r . Clearly describe the intuition behind this estimator.

Hint: Note that you have two moment conditions involving r .

7. **(22 points - 26 minutes)** Let $X_1, \dots, X_n$ be a random sample from distribution with density function given by

$$f(x) = \frac{x}{\theta^2} e^{-x/\theta}, \quad 0 < x < \infty, \text{ and } 0 < \theta < \infty.$$

For this random variable $E(X) = 2\theta$ and $Var(X) = 2\theta^2$.

- (a) Show that the MLE estimator for θ equals $\bar{X}/2$ where $\bar{X}$ is the sample average.
Advice: Do not overspend your time here, ensure you clarify and annotate the steps required. You do not need the derivations to answer the rest of this question.
- (b) Discuss whether the MLE estimator for θ is unbiased and/or consistent. Explain what we understand by concepts of unbiasedness and consistency.
- (c) What would be a good approximation of the sampling distribution of the MLE estimator for θ ? What is the relevance of this result?
Hint: You may assume that regularity conditions are satisfied so that we can rely on the CLT

TABLE FD.1 Observations on Income and Education, 20 Observations

$I =$ Observation,

$$Y = \text{Income.}$$

Source: Data are artificial.

APPENDIX G



STATISTICAL TABLES

TABLE G.1 Cumulative Normal Distribution. Table Entry Is $\Phi(z) = \text{Prob}[Z \leq z]$

[illegible]